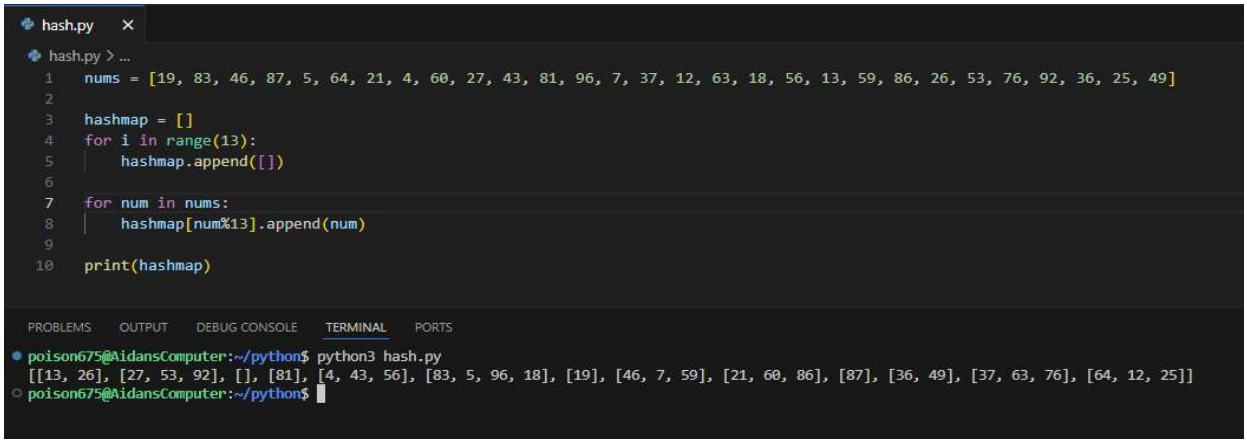
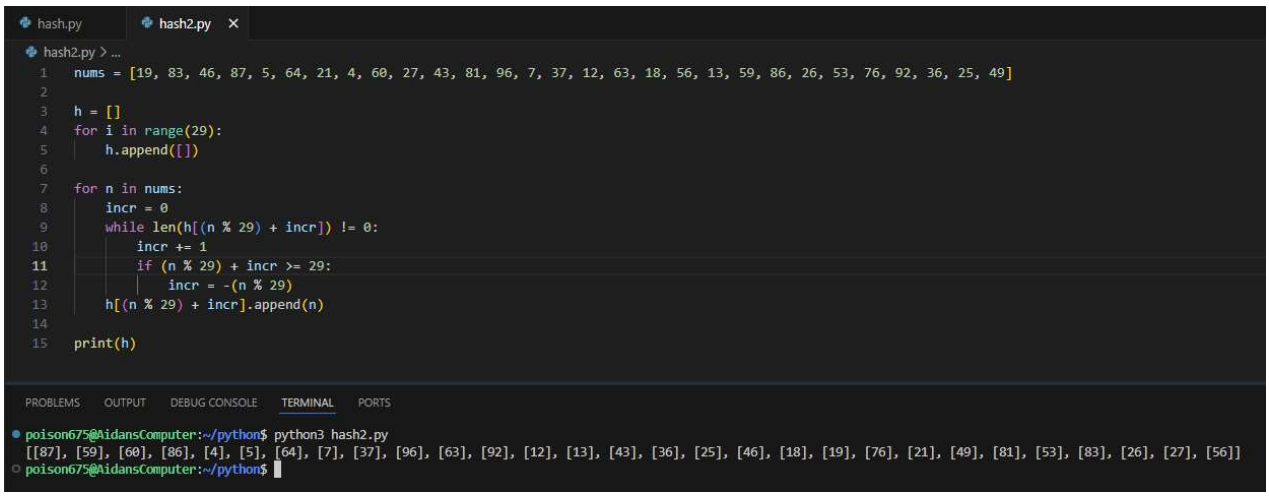
Exercises Week 10

Thursday, October 24, 2024 1:06 PM



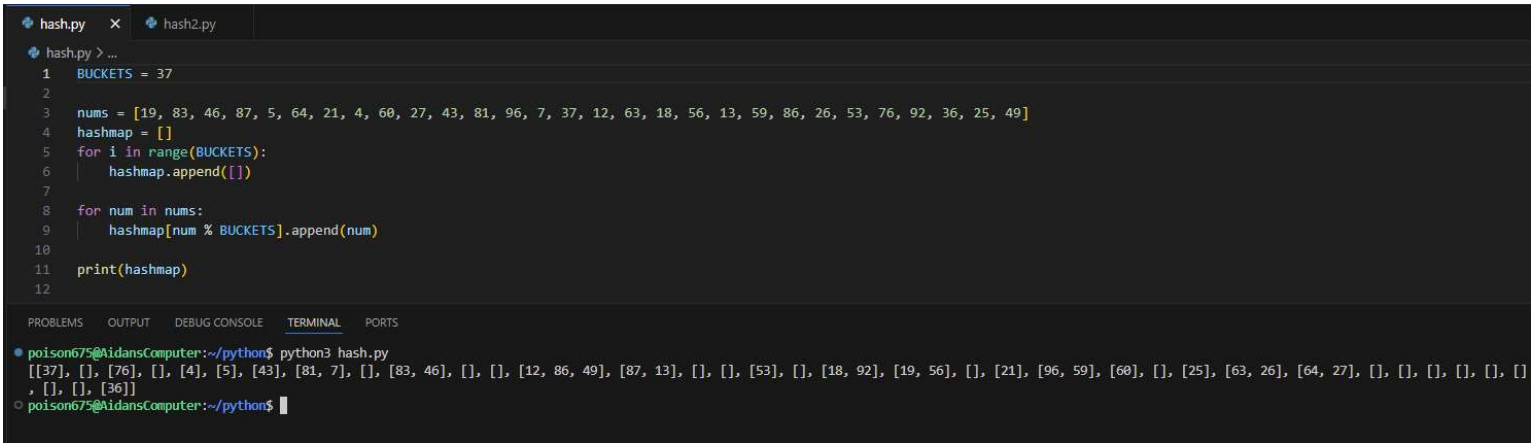
Average search time is based on the distribution of the keys throughout the table. If its a perfect distribution, it will be around (# buckets) ÷ 2 or 6.5.

To find the average search time for this function, the best thing I could come up with is adding up the search times for each element and then dividing by the number of elements. For each element at the beginning of a bucket, it's search time 1, and for subsequent elements, it takes +1 more searches to reach them for how deep they are in the bucket. The average is then: 1.90



Your definition of the load factor doesn't make sense in your notes. You say the load factor is N/n, but also on page 15 that if you choose a large number of buckets "(to reduce the load factor, to an extreme)" which would indicate that it's not N/n, but instead n/N. Also "if the load factor gets really big (n >> N) (n way bigger than N)" which also indicates the load factor is n/N.

That being said, lets choose a load factor of 80% since it seems like you indicated that was a good idea. If you divide the number of entries by 0.8, the next closest prime number is 37, and so choosing to have 37 buckets, this is the result:



The average search time for this is 1.38 operations.

To use a string as a key, simply use any method to turn it into a number. You could use a mod 26 system, where each character represents the number 1-26, or you could convert each letter to mod 10 and then string them together, etc.